Employee Attrition Model Evaluation

AI/ML Internship Task 3 | Model training, evaluation, and responsible-use review

Objective and methodology

This report compares four baseline classifiers and a GridSearchCV-tuned Random Forest on the IBM HR Analytics employee dataset. Task 2's fixed stratified 80/20 split (seed 42) is reused. All preprocessing is cloned and fitted within each training fold during cross-validation, then fitted on the full training partition for final test evaluation. The target Attrition is never included in model features.

Dataset and target

The cleaned Task 2 dataset has 1,470 rows and 37 columns; Attrition is the Yes/No target and Yes is positive. The fixed 80/20 split has 1,176 training and 294 test records (247 No, 47 Yes); Task 2 quality checks found no missing cells or exact duplicates.

Algorithms and settings

Held-out test comparison

Positive-class metrics use Attrition=Yes. Test set: 294 records; 47 positive.

Candidate	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.786	0.402	0.702	0.512	0.814
Decision Tree	0.769	0.322	0.404	0.358	0.617
Random Forest (baseline)	0.827	0.389	0.149	0.215	0.786
K-Nearest Neighbors	0.847	0.625	0.106	0.182	0.655
Random Forest (GridSearchCV tuned)	0.833	0.480	0.511	0.495	0.792

Grid search and model selection

5-fold stratified GridSearchCV used F1 scoring on training data only. Best parameters: {'classifier__max_depth': 5, 'classifier__min_samples_split': 5, 'classifier__n_estimators': 200}; mean CV F1: 0.485. Selected candidate: Logistic Regression by training-only mean CV F1. The held-out comparison is evaluation-only and did not participate in candidate selection.

Business interpretation

The target is imbalanced. Accuracy alone can conceal missed attrition cases. Higher recall may identify more employees for supportive conversations but can also create false alarms; higher precision lowers unnecessary outreach but can miss employees needing support. ROC-AUC summarizes ranking across thresholds; it is not a deployment threshold. Correlation and feature importance are not causal evidence.

Responsible use and limitations

The data is a benchmark, not a causal or employer-specific study. Small positive class, historical bias, proxy features, and distribution shift limit generalization. Never use scores for punitive or automated employment decisions. Minimize data access and retention; assess subgroup error rates and disparate impact; use human review and offer fair support independently of a score.

Reproducibility and persistence

Random seed 42; Task 2 fixed split reused (1,176 train / 294 test). Preprocessing is inside the serialized sklearn Pipeline and never fit on test data. Saved model reloaded with identical predictions on all 294 test rows; 5 unseen compatible rows including an unknown JobRole were accepted.

ROC curves

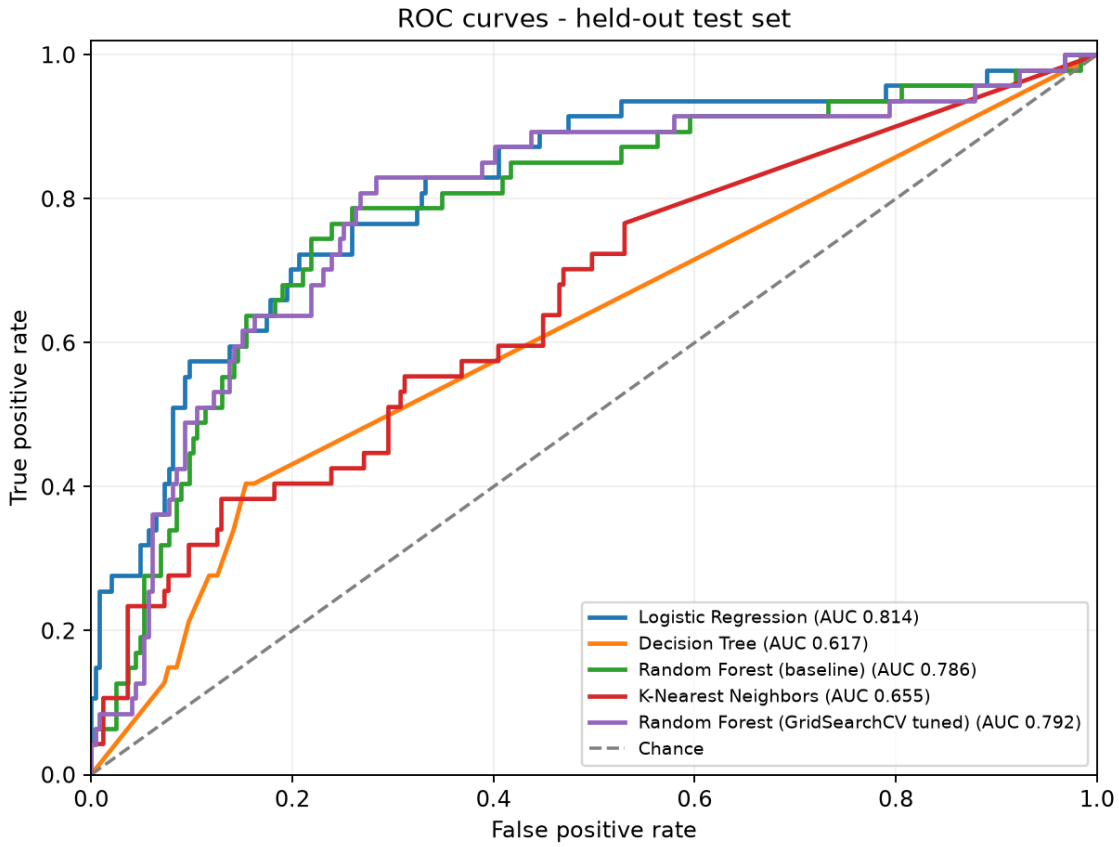


Figure 1. ROC curves generated from held-out test probabilities.

Feature importance

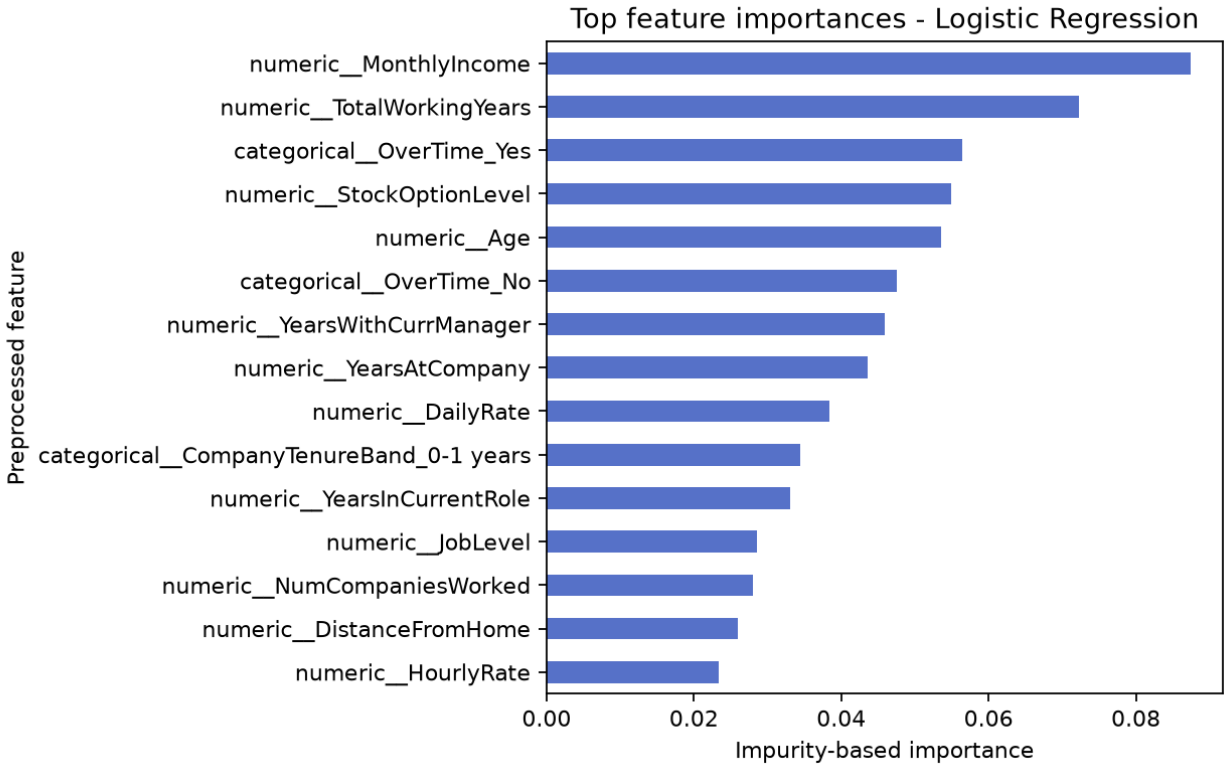


Figure 2. Top impurity-based feature importances for the tuned Random Forest; this is not causal evidence.

Confusion matrix: Logistic Regression

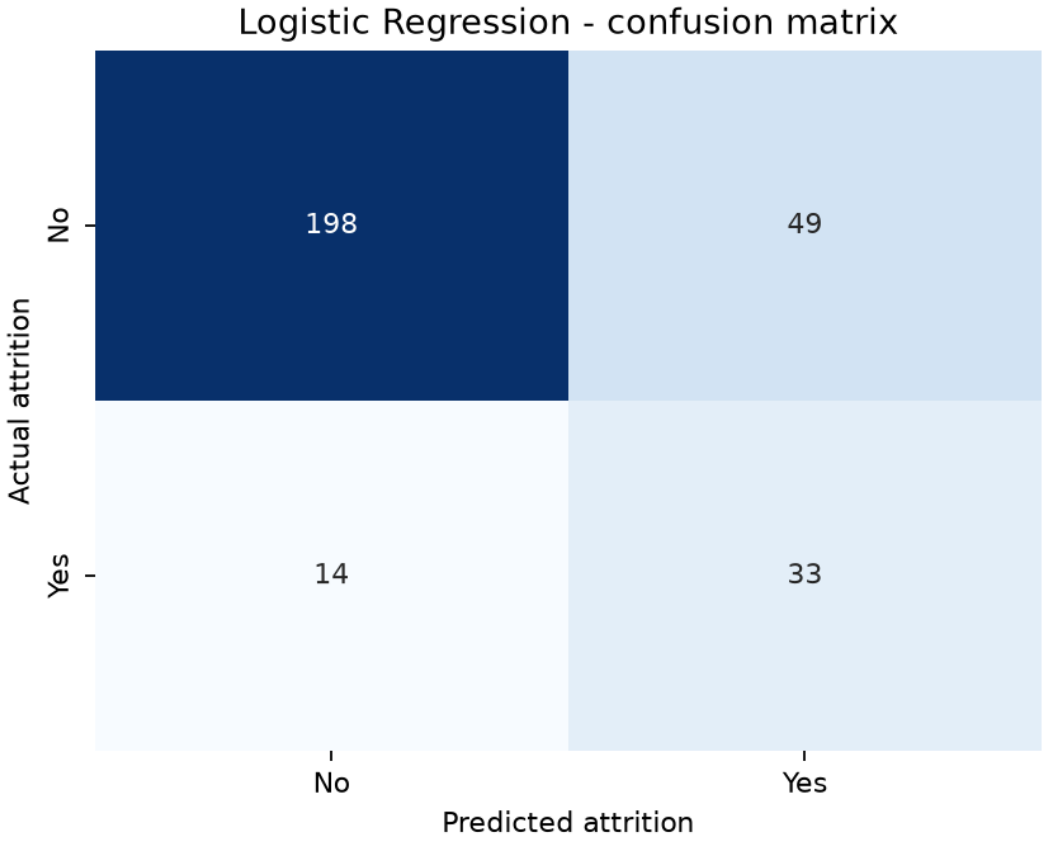


Figure. Rows are actual labels; columns are predicted labels. Attrition=Yes is the positive class.

Confusion matrix: Decision Tree

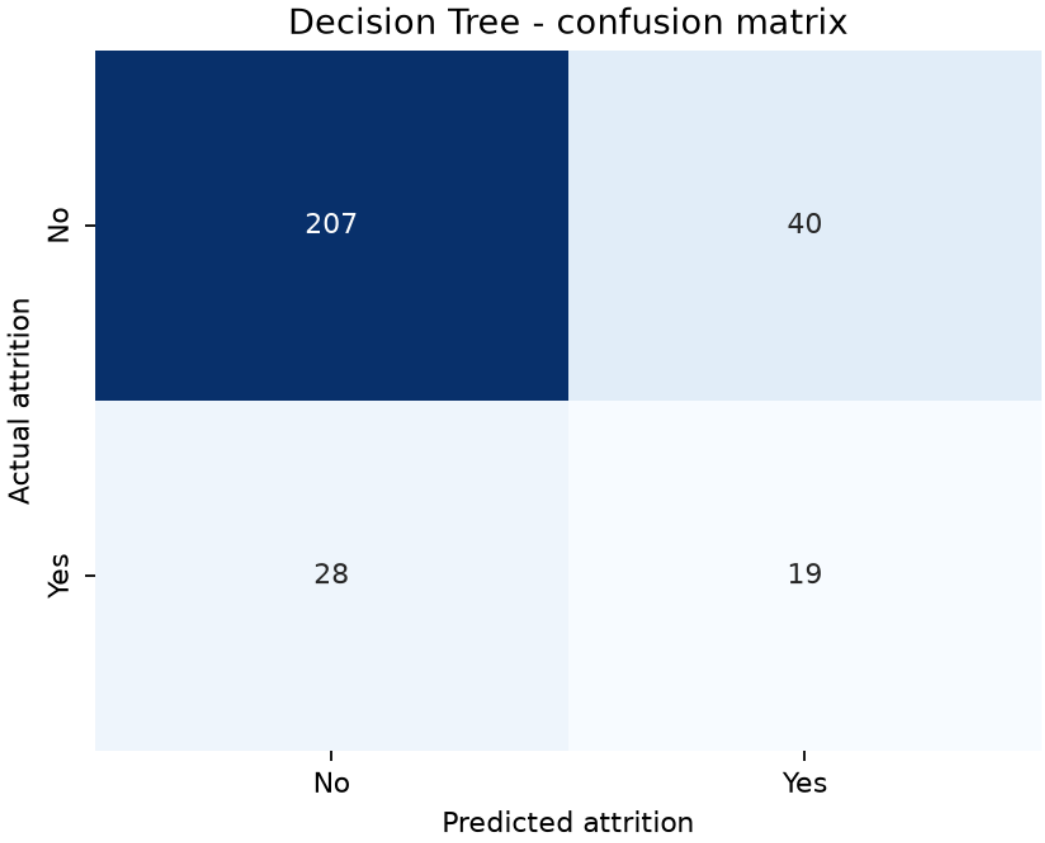


Figure. Rows are actual labels; columns are predicted labels. Attrition=Yes is the positive class.

Confusion matrix: Random Forest (baseline)

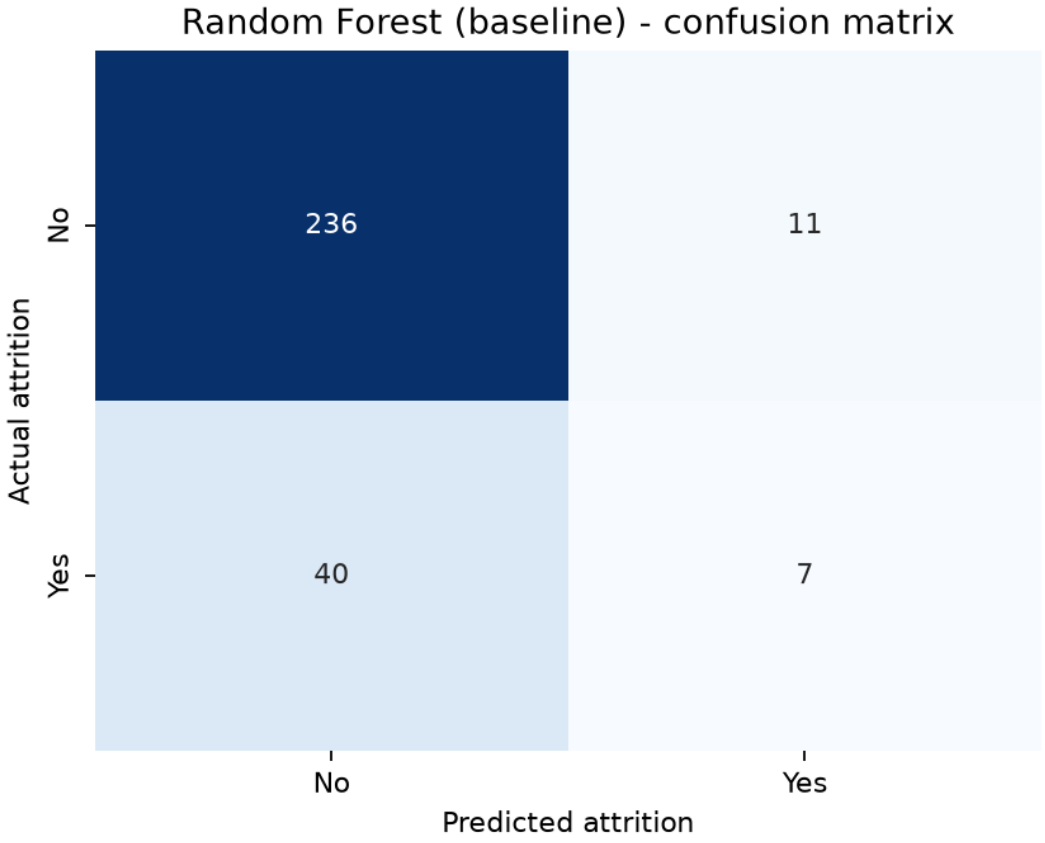


Figure. Rows are actual labels; columns are predicted labels. Attrition=Yes is the positive class.

Confusion matrix: K-Nearest Neighbors

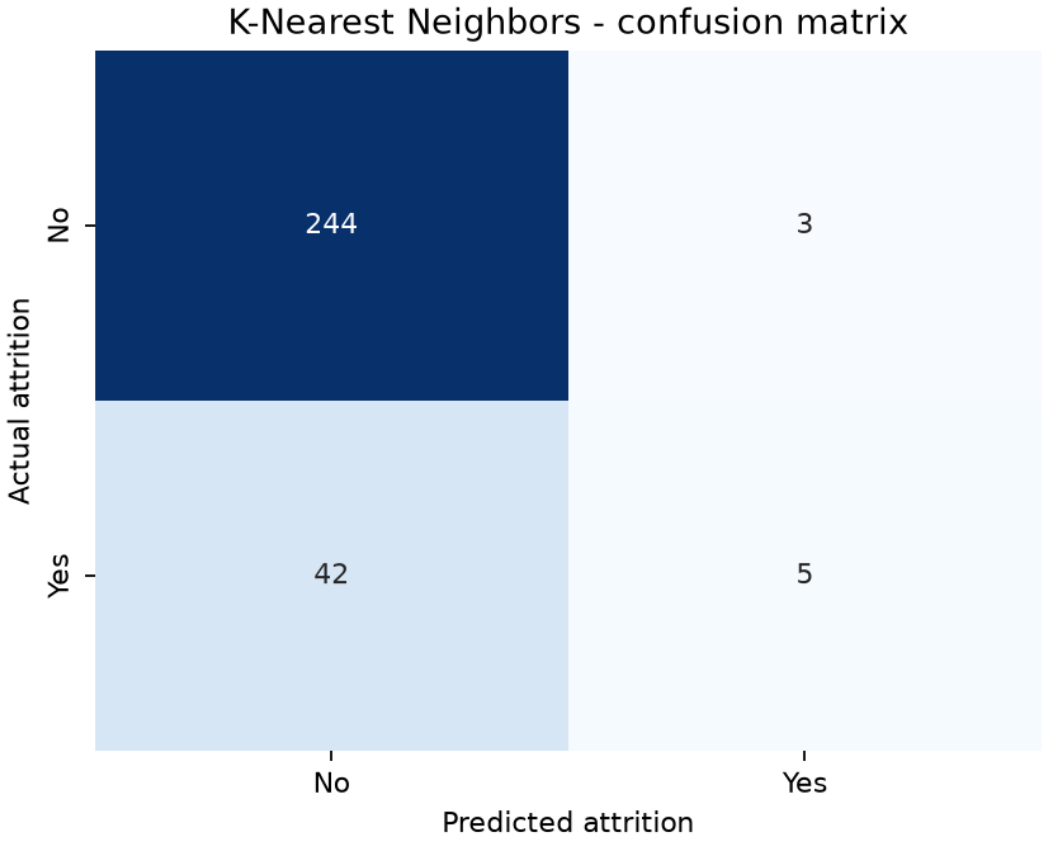


Figure. Rows are actual labels; columns are predicted labels. Attrition=Yes is the positive class.

Confusion matrix: Random Forest (GridSearchCV tuned)

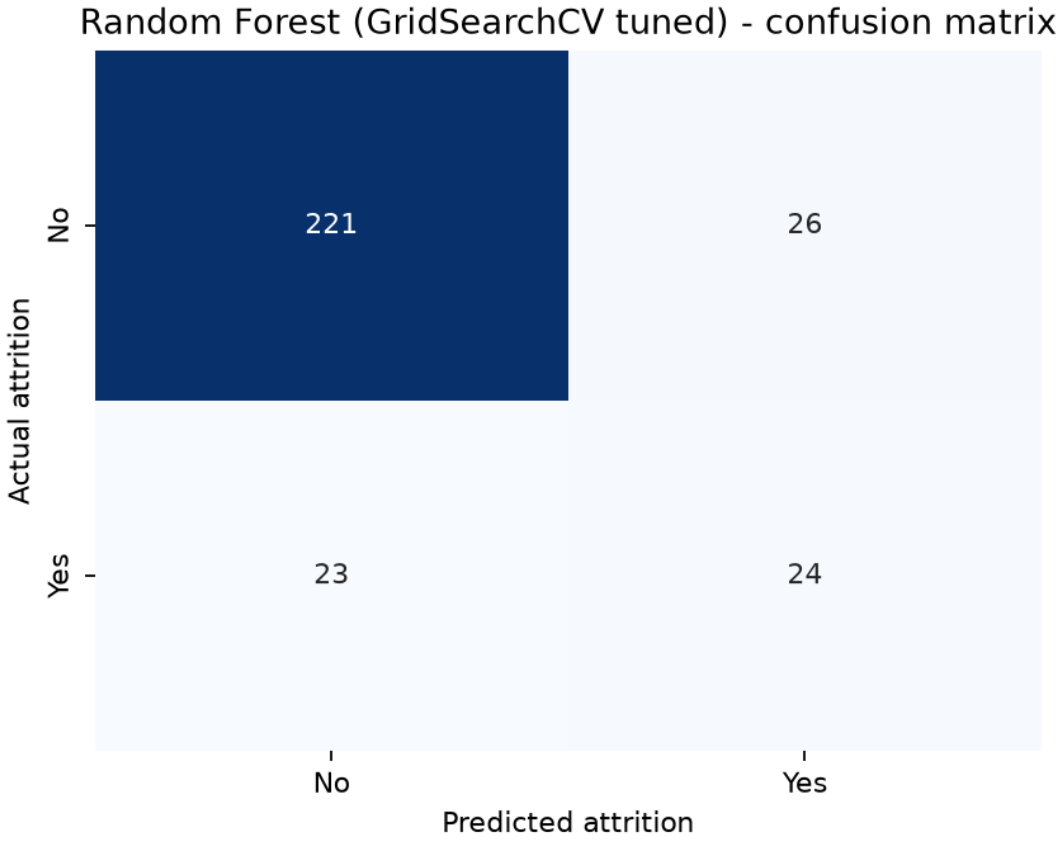


Figure. Rows are actual labels; columns are predicted labels. Attrition=Yes is the positive class.

References

IBM HR Analytics Employee Attrition & Performance dataset (educational benchmark); https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset. Scikit-learn user guide: Pipeline, ColumnTransformer, GridSearchCV, model evaluation.